

BRIDGEPULSE · REQUIREMENTS

BridgePulse requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bridgepulse		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	R9 states the tamper-resistant fixing chosen on 2026-10-02 (crimped stainless banding); FieldNode pinout recorded as decided; no status changed
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; R12 status reworded, no number changed
0.5	2026-09-30	Amish Chadha	Status from BRP-CAL-001 v0.3 after the design for construction (BRP-DDR-004); R12 now not met by \$2.00; no target changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R2 method (load gating), R5 verification, R7 restated with an 11-byte reduced summary; status from BRP-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	R12 redefined per BRP-DDR-001 D1; status of every requirement from BRP-CAL-001; interface assumptions for FieldNode
0.2	2026-09-25	Amish Chadha	

First measurable requirements for TRL 2, with status against the concept estimates

0.1	2026-09-25	Amish Chadha	Initial scaffold
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BridgePulse requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with bridge owners or inspectors, and will be revised after co-design sessions (see BRP-PRB-001). Status is now judged by calculation in BRP-CAL-001 against the parametric model and the priced BOM; "met on paper" is not a verification by test.

Reference installation. A steel or concrete footbridge or short-span road bridge of 5 to 30 m span, monitored at midspan, in a climate from -20 to $+50$ °C, with LoRaWAN coverage and no mains power. The worked example is the 7 m steel footbridge in BRP-PRC-001.

Changes in v0.3. R12 is redefined to cost the BridgePulse-specific parts, with the FieldNode core costed in its own repo and the complete cost stated beside it (BRP-DDR-001, D1; decided by Amish, 2026-09-25: go with recommendation). No other target changes.

Changes in v0.5 (BRP-DDR-004). No target changes. With the design made buildable, BRP-CAL-001 v0.3 finds R12 over the value-engineering target by \$2.00; R9 stays met on paper (10 mm below the soffit, the bolts now ending inside the clamp jaws).

Changes in v0.4 (BRP-DDR-002). R2 keeps its target; its method now gates out the time when people are on the span, backed by a load regression in TwinKit, and is to be checked on recorded data before any change to the target. R5 keeps its target; verification adds an ice-point check, and the probe is now TMP1826 class. R7 is restated so that an 11-byte reduced summary is used wherever the data rate in use limits the payload to 11 bytes. Statuses follow BRP-CAL-001 v0.2.

Table 1. Requirements and status at TRL 3 (BRP-CAL-001 v0.3, Table 5).

ID	Requirement	Target	Verification	Status at TRL 3
R1	Measure bridge acceleration	3 axes; noise density 25 $\mu\text{g}/\text{VHz}$ or less; band 0.5 to 60 Hz	Datasheet; later bench noise test	Met by design (22.5 $\mu\text{g}/\text{VHz}$; 62.5 Hz filter corner)
R2	Track natural frequencies	Every mode with a clear peak below 60 Hz, at least the first vertical mode; bin width 0.02 Hz or less per hourly record; repeatability 0.2 % (1σ) or better over a day of steady temperature	Spectral and load-gating simulation (BRP-CAL-001, B); then recorded data	At risk: with the time when people are on the span gated out, 0.015 % or less in simulation (1 to 20 crossings per record); ungated, walker mass gives 1.5 to 3.1 %
R3	Flag structural change	Flag a sustained shift of 1 % or more in a tracked frequency, after temperature compensation, within 14 days; one false flag per bridge per year or fewer	Detection statistics (BRP-CAL-001, C); then public benchmark data and a field baseline	Not verifiable at TRL 3: needs a daily residual of 0.75 % or less after temperature and load compensation
R4	Measure strain	2 channels; range $\pm 1,000 \mu\epsilon$; resolution 2 $\mu\epsilon$ or better; drift 5 $\mu\epsilon$ per month or less after temperature compensation	ADC noise calculation; later outdoor drift test	Not verifiable at TRL 3 for drift; resolution (0.09 $\mu\epsilon$ RMS) and range met on paper
R5	Measure temperature	Steel surface and shaded air; $\pm 0.5 \text{ }^\circ\text{C}$ or better, checked against a reference	Datasheet; ice-point check at 0 $^\circ\text{C}$ and CalRig chamber check	Met on paper: TMP1826 class, $\pm 0.3 \text{ }^\circ\text{C}$ from -40 to $+105 \text{ }^\circ\text{C}$ on the datasheet check
R6	Stay within the FieldNode energy budget	Average draw 30 mW or less at the FieldNode sensor port	Power calculation; later current logging	Met on paper (16.9 mW)
R7	Send and keep the data	One summary per hour of 48 bytes or less, within LoRaWAN regional limits at the data rate in use, with an 11-byte reduced summary (first frequency, its amplitude, two strain means, steel temperature, status) wherever the limit is 11 bytes; raw records kept on site for 12 months or more	Payload and airtime calculation; storage calculation	Met on paper: 36 bytes, or 11 bytes at US915 DR0 and AS923 DR2 (AS923 limit assumed); 22 months on 32 GB
R8	Fit without harming the structure	No drilling, welding or cutting; strain gauges only on an area of paint removed and recoated with the owner's permission; fitted by two trained people in 4 h or less from below or beside the deck	Fitting sequence estimate; later trial fit	Not verifiable at TRL 3 for time (2.8 h estimate); no drilling or welding met by design
R9	Keep clear of what passes under and over the bridge	Nothing more than 15 mm below the girder soffit; nothing on the deck side except the FieldNode on a handrail post, with tamper-resistant fasteners (one-way crimped stainless banding on installed units)	Model check	Met on paper (10 mm; hub inside the girder outline)
R10	Survive outdoors	Hub IP67, FieldNode IP65; operate from -20 to $+50 \text{ }^\circ\text{C}$; 5 years with one gauge recoat and no cell replacement	Datasheets and design review; later environmental checks	At risk: FieldNode heat (FND-CAL-001) and gauge coating life

ID	Requirement	Target	Verification	Status at TRL 3
R11	Protect privacy	No cameras or microphones; only structural readings leave the device	Design review	Met by design
R12	Affordable	BridgePulse-specific parts within the \$250 value-engineering target (a hypothetical control target), with the complete cost including the FieldNode core stated beside it	Priced BOM (<code>bom/</code> <code>bom.csv</code>)	Over the value-engineering target by \$2.00: \$252.00 estimated; complete \$378.00
R13	Present results responsibly	Every output labeled as monitoring data that supports inspection; no pass or fail safety rating anywhere in the system	Review of dashboard and report templates	Met by design intent; not yet implemented

■ Requirements not met or at risk

- **R12 (cost) over the value-engineering target by \$2.00:** the parts added to make the design buildable (BRP-DDR-004: two probe glands, foot blocks, jaws, packers, jack block and screws) bring the estimated cost of the BridgePulse-specific parts to \$252.00 on indicative prices, against the \$250 target. The target is a hypothetical control target, not a limit; savings worth trying are in the value engineering section of BRP-DEC-001.
- **R2 (repeatability) at risk:** on a light footbridge the mass of each walker lowers the first mode by up to 6.3 %. Gating out the time when someone is on the span meets 0.2 % in simulation, but the footfall model is idealized and the method must be checked on recorded data (TRL 4, on hold).
- **R3 (change detection) not verifiable at TRL 3:** it depends on modeling temperature and any remaining load effect to a daily residual of 0.75 % or less.
- **R10 at risk:** FieldNode heat and coating life.
- **R4 and R8 not verifiable at TRL 3:** gauge drift and fitting time need tests or a trial fit.

■ Assumptions

- The accelerometer follows the ADXL355 datasheet summary ([Analog Devices](#)); a different part must meet R1.
- The FieldNode core offers 100 mW average for sensors as its design value (115 mW ceiling, FND-CAL-001) and two sealed M12 5-pin ports with a switched rail. BridgePulse uses RS-485 (Modbus RTU) and a 5 V rail on FieldNode's candidate pinout, adopted as the shared standard on 2026-10-02 (BRP-DEC-001).
- A temperature-compensated frequency baseline needs at least four weeks of data before any flag, following the approach of [Peeters and De Roeck \(2001\)](#).
- Steel modulus 210 GPa for converting strain to stress. The other calculation assumptions are in BRP-CAL-001, Table 1.